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Structured expert judgement projections of the ice-sheet contribution to future sea-level rise

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Key Points:

- A panel of 21 ice-sheet experts estimated that ice sheets will likely contribute −0.2 to 0.8 m to global sea levels by 2100 under 2°C warming, and −0.3 to 1.8 m under 5°C
- Compared to a 2018 study, experts project narrower Greenland ice sheet uncertainty and larger potential increases in Antarctica accumulation
- At the 95th percentile, under SSP2-4.5, ice sheets could contribute 1.1 m by 2100 and 2.5 m by 2150 to global mean sea-level

Abstract

Structured expert judgement (SEJ) is an established method for estimating the likelihood of deeply uncertain outcomes, such as century-to-multicentury ice-sheet changes. We used SEJ to estimate ice-sheet changes through 2300 under scenarios stabilizing in 2100 at 2°C, 3.5°C, and 5°C of global warming. Under a 2°C scenario, the projected median (5th–95th percentile) ice-sheet contribution to GMSL change is 23 (−16 to 76) cm by 2100 and 131 (−45 to 420) cm by 2300; under a 5°C scenario, these values rise to 42 (−34 to 181) cm by 2100 and 365 (10 to 1120) cm by 2300. Relative to a comparable 2018 study, upper-end projections are similar while low-end projections reflect a greater probability of a significant increase in Antarctic accumulation. Combining elicitation results with probabilistic interpolation indicates a GMSL contribution from the Greenland ice sheet under SSP2-4.5 of 16 (−4 to 43) cm by 2100, 33 (−6 to 93) cm by 2150, and 77 (−17 to 234) cm by 2300, exceeding corresponding Intergovernmental Panel on Climate Change Sixth Assessment Report (AR6) medium-confidence estimates in uncertainty and, for 2100 and 2150, median values. For Antarctica, projections of 11 (−34 to 85) cm by 2100, 26 (−54 to 191) cm by 2150, and 105 (−84 to 532) cm by 2300 are more uncertain than AR6 medium-confidence projections. Accounting for tail dependence, 95th-percentile total ice-sheet contributions reach 1.1 m by 2100 and 2.5 m by 2150 under SSP2-4.5, highlighting the potential for multi-meter GMSL rise in the 22nd century under intermediate emissions.

Plain Language Summary

Predicting how much ice-sheet mass loss will raise global sea levels over the coming centuries is difficult because complex ice-sheet processes remain deeply uncertain. To address this, we used Structured Expert Judgement (SEJ)—a process that calibrates and pools expert predictions—to combine the estimates of 21 ice-sheet scientists on future changes to the Greenland and Antarctic ice sheets under 2°C, 3.5°C, and 5°C warming scenarios through the year 2300. Under a low 2°C warming scenario, experts project ice loss will add a median of 0.2 m (range of −0.2 to 0.8 m) to sea levels by 2100 and 1.3 (−0.5 to 4.2) m by 2300. Under a severe 5°C warming scenario, those estimates jump to 0.4 (−0.3 to 1.8) m by 2100 and 3.7 (0.1 to 11.2) m by 2300. While high-end projections remained similar to the findings of a 2018 SEJ study, the lower-end estimates dropped because experts accounted for increased snowfall accumulating on Antarctica. The experts expressed conflicting positions on whether and how to include marine ice-cliff instability. When applied to an intermediate emissions pathway (SSP2-4.5), the study's projections for both Greenland and Antarctica are higher and carry greater uncertainty than the likely range estimates from the IPCC Sixth Assessment Report (AR6). Overall, the findings show a 5% chance under this scenario that ice loss alone could contribute 1.1 meters to global sea levels by 2100 and 2.5 meters by 2150—underscoring that multi-meter sea-level rise remains a serious risk for future centuries even under moderate emissions.

1 Introduction

Sea-level rise is among the most enduring effects of climate change, and for the approximately 1 billion people living within 10 m of sea level, among the most impactful (Seeger & Minderhoud, 2026). Beyond the next few decades, it is also one of the most uncertain, due to the dominant role of complex, multi-scale ice-sheet processes influencing high-end sea-level outcomes (e.g., Fox-Kemper et al., 2021). This uncertainty has been recognized throughout the five decades of modern sea-level rise projection efforts (e.g., Gornitz et al., 1982), but nonetheless remains. Indeed, the response of ice sheets to climate change on century-to-multicentury timescales is a canonical example of deep uncertainty, defined as a situation in which "analysts do not know, or the parties to a decision cannot agree on, (1) the appropriate models to describe the interactions among a system's variables, (2) the probability distributions to represent uncertainty about key variables and parameters in the models, and/or (3) how to value the desirability of alternative outcomes" (Kopp, Oppenheimer, et al., 2023; Lempert, 2002). In this paper, we refer to elements 1 and 2 of the definition of deep uncertainty as ambiguity.

Even under conditions of deep uncertainty, some decision frameworks and risk communication approaches rely upon expressing uncertainty in terms of probability distributions. One well-established approach to generating probabilistic estimates under these circumstances is structured expert judgement (SEJ) (e.g., Cooke, 2018; Cooke et al., 2021). SEJ provides a formal method of combining input from experts in a calibrated manner to estimate uncertain quantities. In the 'Classical model', experts first quantify uncertainty by assessing 5th, 50th- and 95th-percentiles for calibration questions from their field for which the true values are known but not immediately accessible. The informativeness and accuracy of their answers are used to weight each expert's responses to target questions whose answers are the objects of the study.

SEJ has been applied multiple times to estimate the ice-sheet contribution to future sea-level change. Spurred in part by the Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report (AR4)'s refusal to assess the likelihood of future ice-sheet changes (Meehl et al., 2007), Bamber and Aspinall (2013) assessed the rate of West Antarctic ice sheet (WAIS),

East Antarctic ice sheet (EAIS), and Greenland ice sheet (GrIS) mass loss in 2100 based on experts' expectations regarding future climate. They found a median ice-sheet contribution to GMSL over 2010–2100 of 29 cm (90% CI of 10–84 cm). Bamber et al. (2019) describe a 2018 elicitation (henceforth, 2018 SEJ) of contributions to global-mean sea-level (GMSL) change from both discharge and surface-mass balance (SMB) through 2300, under 2°C and 5°C stabilization scenarios. They found a median 21st century ice-sheet contribution to GMSL of 26 (3–81) cm under a 2°C stabilization scenario, and of 51 (7–178) cm under a 5°C scenario.

The 2018 SEJ results were noted in the IPCC's Special Report on Oceans and Cryosphere in a Changing Climate (Oppenheimer et al., 2019) and played an important role in IPCC's Sixth Assessment Report (AR6) (Fox-Kemper et al., 2021). In particular, the 2018 SEJ results served as one of the key lines of evidence for quantitative representations of high-impact ice-sheet processes with poorly-known likelihood, as well as their incorporation into 'low confidence' sea-level rise projections illustrating the potential consequences of these processes (Kopp, Oppenheimer, et al., 2023). They helped lead AR6 to conclude that "Global mean sea level rise above the likely range—approaching 2 m by 2100 and 5 m by 2150 under a very high greenhouse gas emissions scenario (SSP5-8.5) (*low confidence*)—cannot be ruled out due to deep uncertainty in ice-sheet processes" (IPCC, 2021).

Substantial advances in the understanding of ice-sheet behavior have occurred since the 2018 SEJ. First, the observational record, which is most comprehensive from 1992, is extended by an additional seven years, providing a roughly 30 percent longer time series of more than three decades. Because ice sheet dynamics operate over a wide range of timescales from diurnal to millennial (Bamber et al., 2007) and because inter-annual variability in SMB is large (Otosaka et al., 2026; Wouters et al., 2013), a longer observational time series is a valuable asset in understanding contemporary ice sheet behavior. Here, the extended time series does not support a sustained acceleration in mass loss, which would require a longer time series to identify robustly (Wouters et al., 2013). Instead, it suggests a step change in the late 1990s/early 2000s for the WAIS and GrIS with no trend for the EAIS and large inter-annual variability (Otosaka et al., 2026; Stokes et al., 2025).

Second, accordingly, research has been directed at addressing the critical question of how much of the recent trends in imbalance observed for the WAIS and GrIS have been driven by external forcing versus internal variability (e.g., Haigh & Holland, 2024; King et al., 2023; Naughten et al., 2023).

Third, advances have been made in modelling individual processes as well as fully coupled ice sheet/global climate model simulations (e.g., Siahaan et al., 2022). In particular, the role of Marine Ice Cliff Instability (MICI) has been extensively studied, especially for the WAIS. While its role is still contested, multiple modeling studies suggest that its significance in the 21st century may be limited (e.g., Clerc et al., 2019; DeConto et al., 2021; Morlighem et al., 2024).

In light of these advances, this study updates Bamber et al. (2019), with the addition of an intermediate (3.5°C) trajectory. It builds upon the SEJ literature in two important ways. First, it leverages an innovation introduced in the 2018 SEJ: assessing dependences and tail dependences between ice sheet processes (Kurowicka et al., 2024). Using the vine-copula method (Nagler et al., 2026), we applied the 2018 SEJ dependence structure to the current assessment, allowing this study to focus on other changes in the literature. Second, we leverage advances in online expert elicitation, adapting training modules from the World Health Organization (Lake et al., 2026) to apparent good effect.

Compared to the 2018 SEJ, we find that experts elicited in a 2025 panel (henceforth, 2025 SEJ) are more skilled in evaluating their own uncertainty, as reflected in performance on calibration questions; are more certain in their projections of 21st century changes in the GrIS; and put greater weight on the possibility of a substantial increase in Antarctic accumulation, thereby lowering 5th percentile projections of ice sheet mass loss. Overall, median and 83rd percentile SEJ projections for GrIS mass loss in the 21st century are higher than in the set of IPCC AR6 projections that incorporate only processes in whose dynamics there is at least medium confidence, while 2300 SEJ projections for GrIS are more uncertain (Fox-Kemper et al., 2021). Twenty-first century AIS SEJ projections are more temperature-dependent than the medium confidence projections in AR6, while also more uncertain at both the low and the high end; 2300 projections are similarly more uncertain (Fox-Kemper et al., 2021). Consistent with

Bamber et al. (2019), 95th percentile ice-sheet contributions to GMSL approaching 1.2 m under a 3.5°C stabilization scenario and 1.8 m under a 5°C stabilization scenario indicate the total 21st century GMSL rise approaching 2 m remains within the realm of possibility. 95th percentile ice-sheet projections of 3.6 m GMSL for 2200 under a 3.5°C scenario emphasize the possibility of multiple meters of GMSL rise in the 22nd century.

The remainder of the paper is structured as follows. Section 2 discusses the methods of the analysis. Section 3 describes this paper's contribution to the literature on SEJ, focusing on the improved performance of experts in this study compared to the 2018 SEJ. Section 4 describes and discusses the experts' quantitative projections for ice-sheet change under temperature-based stabilization scenarios, then uses these projections to calibrate projections for ice-sheet change under alternative future emissions scenarios. Section 5 delves into the experts' reasoning underlying their judgements. Section 6 concludes.

2 Methods

2.1 Overview of the Classical Method of SEJ

The current study builds on the SEJ study of Bamber et al. (2019). For an exposition of this method the reader is referred to that study; for deeper background, see Cooke et al. (2021).

In brief, experts quantify their uncertainty both for variables of interest and also for calibration variables from their field, for which true values are available (post hoc) but unknown to the experts. In this study, expert performance was calibrated using a variety of questions related to ice-sheet and climate science. The calibration questions were answered during the online elicitation in which experts are asked to assess their degrees of belief – expressed as three uncertainty values corresponding to their personal judgments of fifth, 50th and 95th percentiles – without consulting external sources (Text S1). Using the calibration questions, expert assessments are scored on Statistical Accuracy (SA) and Informativeness (Inf), with SA weighted more heavily in expert evaluation. SA measures the likelihood that an expert's percentile assessments 'correspond to reality' in the sense that there is a 5% chance of a

calibration variable falling at or below the 5 percentile, a 45% chance of falling above the 5 percentile and at or below the 50 percentile, etc. Inf is a dimensionless measure of the degree to which a distribution is concentrated, relative to a reference background distribution. The narrower an expert's 5–95th percentile range, the more informative they are. An overconfident expert will have high Inf but low SA; one who accurately believes themselves to be ignorant will have low Inf and high SA. One who is both knowledgeable and skilled at quantifying the limits of their knowledge will have high SA, and as high an Inf as is reasonable given those limits.

These scores are used to form performance-weighted combinations of experts' uncertainty. Combinations are termed decision makers (DMs). Performance-weighted and equal-weighted (EW) DMs are evaluated on calibration questions, and a choice is made on the basis of performance and robustness. Performance is measured based on the SA and Inf of the DM. Robustness entails that the combined score of the DM should not change much by excluding a single expert or a single calibration variable (for details, see Text S2–4; Tables S1–S8).

2.2 Target questions

The target questions (i.e., those related to future ice sheet contributions to sea level) asked experts to assess 5th, 50th, and 95th percentile contributions to GMSL change (relative to a 1995–2014 baseline) from three processes—accumulation, runoff, and discharge—for each of WAIS, EAIS and GrIS, at three epochs (2100, 2200, and 2300), under three global-mean surface temperature scenarios (rising from 1.2°C above the late-nineteenth century in 2020 to 1.7°C, 2°C, and 2.5°C in 2050, then continuing to warm to 2°C, 3.5°C, and 5°C in 2100, and subsequently stabilizing) (Figure 1). Target questions are provided in Text S1. To supply the experts with a reference point for checking their answers, the experts were given estimates of decadal-mean contemporary rates for each process (Otosaka et al., 2023) and what these values would imply for GMSL if they continued at a constant rate to 2100, 2200, and 2300.

Summing expert assessments of individual processes into estimates of total change for each ice sheet, as well as the total across ice sheets, requires an evaluation of the dependence structure. For example, it requires answers to questions such as: How does accumulation

correlate with runoff? Does WAIS mass loss correlate with EAIS mass loss? In this study, at each time point, elicited accumulation, runoff, and discharge projects were combined into total changes in each ice-sheets and across ice sheets using the Bamber et al. (2019) elicited dependence structure (see Text S5; Table S9; Figure S1).

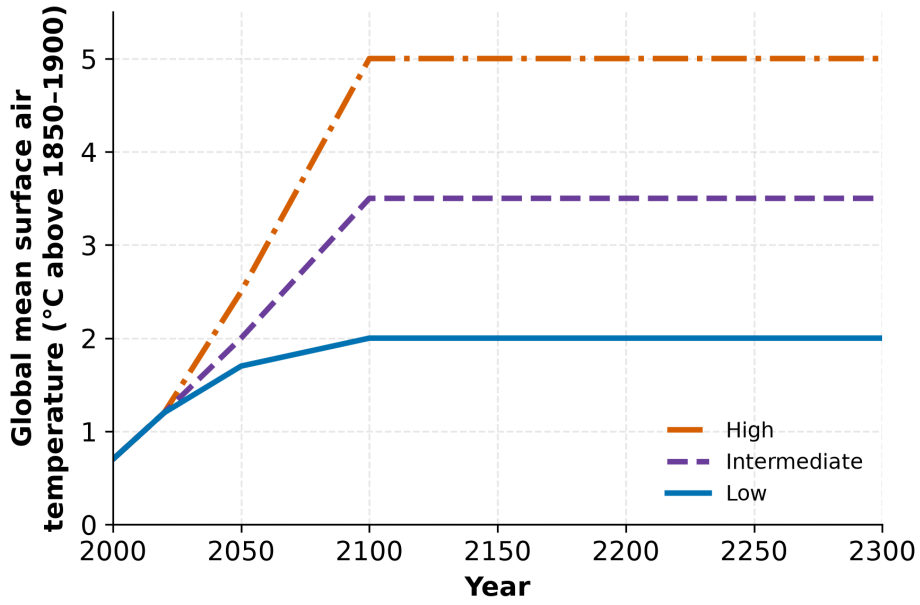


Figure 1. Scenarios of global mean surface air temperature used in this study.

2.3 Recruitment and structure

Twenty-one experts participated in the 2025 SEJ, eight of whom had also participated in the 2018 SEJ. The elicitation involved two live, online, video plenary discussions in October 2025 to clarify the method. During the online sessions, the experts also individually answered the calibration questions. In answering these questions, the experts were asked to assess their degrees of belief – expressed as three uncertainty values corresponding to their personal judgments of fifth, 50th and 95th percentiles – without consulting external sources. The experts then answered the target questions individually offline, mostly within two weeks after the plenary sessions. Responses to both the calibration and target questions were collected via online surveys managed with the Qualtrics survey marker tool. Experts were allowed to choose

whether to assess only GrIS, only AIS, or both. Of the 21 experts, three assessed only GrIS, three assessed only AIS, and 15 assessed all ice sheets.

A recent large-scale elicitation for the World Health Organization (Lake et al., 2026) invested heavily in online training tools to support experts' uncertainty quantification, with positive gains in performance (Nane et al., 2026). Taking advantage of that work, the training module was adapted for the current study.

2.4 Narrative Rationales

As part of the elicitation procedure, all 21 participating experts were invited to provide narrative rationales summarizing the key factors and considerations underlying their scientific thinking when assessing quantitative uncertainty on the contributions to sea-level rise (SLR) by ice sheet processes (see Text S6). Specifically, they were asked to provide a description, for each ice sheet and process elicited, of how they considered evolution over time and at different temperature stabilisations. They were asked to focus “on any transitions in behaviour, such as the onset of strong positive feedbacks such as MICI, marine ice sheet instability (MISI) or the lapse-rate elevation feedback, or a shift from linear to exponential response with temperature or time or any other process [they] considered to be important at a certain time, temperature and/or confidence level.” Thirteen of the experts (11 assessing all ice sheets and two assessing only AIS) gave reasoning underpinning their judgments for some or all the process-related items. The narrative rationales were summarized both manually and through an analysis with a large language model (see Text S7). Experts who provided narrative rationales are included on this paper as co-authors.

2.5 Sea-level rise projections under climate scenarios

Experts assessed ice-sheet contributions to SLR in 2100, 2200, and 2300. Integrating these results into usable, probabilistic time series of projected SLR under standard climate scenarios requires two extensions: (1) interpolating temporally so as to create complete time series from 2000–2300; (2) an interpolation or mixture approach to producing reasonable projections for scenarios other than three explicitly elicited.

2.5.1 Time series construction

Ice Sheet Mass Balance Intercomparison Exercise (IMBIE; Otosaka et al., 2026) observations from 1979–2023 were used to define historical ice-sheet trajectories and the rate of change at the end of the observational record.

For 2050, we used model-derived distributions of ice-sheet contributions from the emulandice module of the Framework for Assessing Changes to Sea-level (FACTS) (FACTS2 Development Team, 2026; Kopp, Garner, et al., 2023). The emulandice module is based on Edwards et al. (2021) and uses Gaussian-process emulators trained on Ice Sheet Model Intercomparison for CMIP6 (ISMIP6) ice-sheet simulations (Goelzer et al., 2020; Seroussi et al., 2020).

The emulandice distributions were used to define a 2050 projection knot for each ice sheet, providing intermediate constraints between the observational record and longer-term SEJ projections.

We imposed temporal dependence across the 2050, 2100, 2200, and 2300 projections using scenario-specific correlation structures derived from DeConto et al. (2021) simulations for EAIS and WAIS and fitted ISMIP6 projections (Fox-Kemper et al., 2021; Kopp, Garner, et al., 2023) for GrIS (Table S12). DeConto et al. (2021) was used in lieu of ISMIP6 for Antarctica to avoid imposing an overly strong correlation between near-term and long-term mass loss (Kopp et al., 2017). The model-derived 2050 distributions were coupled to the SEJ projections while preserving the marginal distributions at each projection year and imposing the target temporal dependence structure. We generated annual ice-sheet contributions from 2024–2300 by interpolating the historical endpoint, 2050, and SEJ projections constraints with a tension spline

2.5.2 Scenario mixing

We produced global surface air temperature (GSAT) trajectories and corresponding cumulative warming over 2000–2300 for six CMIP6 (SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-3.4-OS, and SSP5-8.5; O'Neill et al., 2016) and seven CMIP7 (low-to-negative, very low, low, medium-to-low, medium, high-to-low, and high; emission scenarios. Trajectories were

created using the Finite Amplitude Impulse Response (FaIR) climate model version 2.1.4 (Leach et al., 2021) as implemented within FACTS.

Each FaIR cumulative-warming trajectory was compared with the three SEJ reference trajectories in terms of cumulative warming. We calculated range-normalized Euclidean distances to each reference trajectory at 2100 and 2200 and converted these distances to scenario probabilities using inverse-distance weighting, such that trajectories closer to a given SEJ reference scenario received greater weight (Figure S2, S3). We then sampled realizations from the corresponding temporally coupled SEJ ensemble, preserving temporal dependence across projection years and among ice-sheet components. Further details regarding temporal structure, temporal infilling, and sample weighting for scenario assignment are included in Text S8.

The resulting ensembles for each CMIP6 and CMIP7 emission scenario represent a probabilistic mixture of the elicited 2°C, 3.5°C, and 5°C scenarios.

3 Expert performance

Seventeen of the 21 experts completed the online expert training in uncertainty quantification prior to completing the elicitation. This training is likely responsible for a substantial improvement in SA compared to the 2018 SEJ: whereas only three of the 20 experts in the 2018 SEJ had SA scores above 0.1, eight of the 21 in the 2025 SEJ do (Figure 2, Tables S3, S4). In both studies, there is a negative correlation between SA and Inf: more informative experts tend to be less accurate. In Figure 2a, experts closer to the upper right corner have higher combined score and better performance on calibration variables. High SA with low Inf (upper-left corner of Figure 2a) is useful if in fact that is the best achievable. Low SA with high Inf (bottom-right corner of Figure 2a) is not a useful trait and reflects consistently overconfident assessments. This overconfidence was substantially more common among the 2018 SEJ experts than in the present study.

Another important performance indicator, not used in the weighting, is the Mean Absolute Percentage Error (MAPE) (Figure 2b). This measures the normalized difference between the expert's median estimates on the calibration questions and the true values

(specifically, the absolute percentage error is the absolute value of realization minus forecast divided by the realization). Consistent with recent analysis of a large number of SEJ studies, experts' MAPEs and SA have significant negative rank correlation: high SA tends to pair with low MAPE (Cooke, 2026). This negative correlation is not clearly present within either the 2018 or 2025 SEJ individually, but is clearly present when the two studies are pooled. In general, experts in the 2025 SEJ are more accurate and have lower MAPE than those in the 2018 SEJ, reflecting a reduction in overconfident responses. Although the training material made no mention of treating the medians as point estimates, the association between high SA and low point estimate error may be an indirect co-benefit of training for SA.

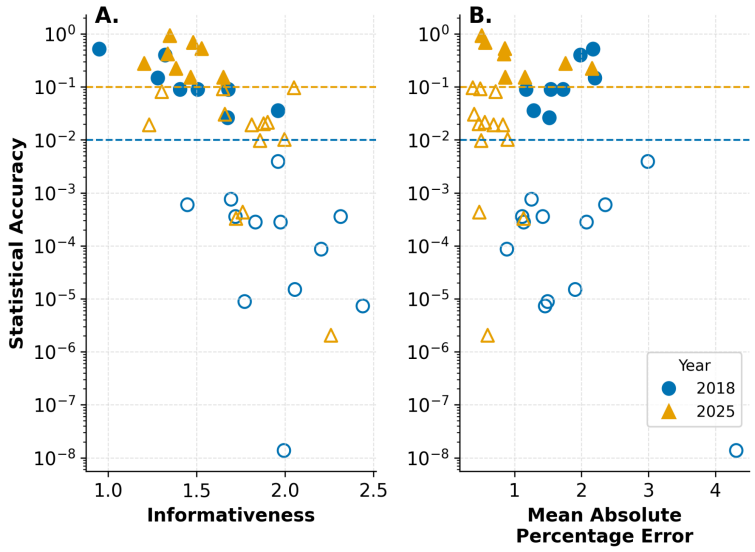


Figure 2. Comparison of the 2018 and 2025 SEJ experts with respect to statistical accuracy versus a) informativeness and b) mean absolute percentage error. Filled in symbols indicate experts with statistical accuracy scores above the statistical accuracy threshold for their respective study (0.01 for 2018; 0.10 for 2025).

Based on combined SA and Inf scores as well as robustness checks (see Text S2–5), the authors selected a SA cut-off of 0.1 for inclusion in the combined DM (henceforth DM I0.1; yellow dashed line in Figure 2). All experts contribute equally to fixing the range, or support, of the distribution, while the weighted experts determine the distribution of probability mass over that range. For DM I0.1, there are six weighted experts in the GrIS panel, six in the AIS panel, and eight in total (the same number as in the 2018 SEJ).

4 Quantitative projections

2.1 Ice sheet contributions to sea-level rise at different warming levels

For GrIS in 2100, the experts project median (17th–83rd percentile) mass loss equivalent to 16 (4–26) cm GMSL rise under the 2°C scenario and 24 (7–43) cm under the 5°C scenario (Tables 1 and S13; Figure 3). While similar to IPCC AR6 projections at the low end, these projections are higher in the median and 83rd percentile than the AR6 median and likely range projections of 8 (4–13) cm for 2.0°C and 15 (11–19) cm for 5.0°C.

Experts project a substantial acceleration in the median and high end over coming centuries, leading to 2300 GrIS projections of 65 (9–137) cm under 2°C stabilization and 118 (39–221) cm under 5°C. Comparison to the AR6 assessed ice-sheet contributions is less direct, since AR6 did not provide 2300 projections based on emissions scenarios rather than warming levels, but in general the AR6 projections are narrower and less fat-tailed. For the low-emissions SSP1-2.6 scenario (which declines from a median projection of about about 2.0°C warming in 2100 to about 1.5°C in 2300), AR6 projects 22–39 cm by 2300; for the very high emissions SSP5-8.5 (which rises from about 5°C in 2100 to about 10°C in 2300), it projects 32–175 cm (Fox-Kemper et al., 2021; Lee et al., 2021).

For AIS in 2100, the experts project 7 (−11 to 32) cm under 2°C and 18 (−14 to 67) cm under 5°C. These projections are dominated by WAIS: under 2°C, the experts give nearly equal odds to mass gain or mass loss in EAIS; in WAIS, they expect net mass loss, while also giving some weight to the possibility of net gain. Compared to the AR6 median and likely range projections for AIS of 11 (3–28) cm under 2.0°C and 12 (4–34) cm under 5°C (Fox-Kemper et al., 2021), the experts give greater credence to both low and high outcomes, while also expecting a substantially greater degree of temperature dependence. Alternative approaches to analyze the ISMIP6 ensemble give slightly greater degrees of temperature dependence than AR6; for example, using gradient boosting emulation of the ISMIP6 ice-sheet multi-model ensemble and incorporating observational constraints, (Lin et al., 2026) project 5 (3–7) cm under the very low emissions SSP1-1.9 and 8 (4–12) cm under SSP5–8.5. Regardless of the point of comparison

with results derived from ice-sheet multi-model ensembles, the expert projections are consistently broader in uncertainty.

This greater uncertainty is even more apparent in 2300, where the expert projections of 59 (−27 to 193) cm under 2°C stabilization and 235 (38–559) cm under 5°C stabilization compared to AR6's assessed ranges of −14 to 78 cm for SSP1-2.6 and −28 to 313 cm for SSP5-8.5 (Fox-Kemper et al., 2021). Notably, however, the AR6 assessment allows for the possibility of net mass gain in AIS through 2300 under SSP5-8.5; the experts allow for that possibility for EAIS (17th percentile of −20 cm) but ascribe a <17% probability to it for AIS as a whole (compare the 17–83rd percentile range of 38–559 cm to the 5th–95th percentile range of −81 to 930 cm).

Table 1. Projected sea-level rise contributions (in cm) from each ice sheet and combined. Values are the fifth, 17th, 50th, 83rd, and 95th percentiles of the Monte Carlo distributions.

Year and ice sheet	2°C			3.5°C			5°C		
	50%	17 to 83%	5 to 95%	50%	17 to 83%	5 to 95%	50%	17 to 83%	5 to 95%
2100									
EAIS	0	−10 to 11	−18 to 20	4	−9 to 20	−20 to 36	4	−12 to 26	−29 to 55
WAIS	6	−4 to 26	−15 to 39	9	−6 to 42	−22 to 64	12	−4 to 48	−24 to 81
GrIS	16	4 to 26	−4 to 35	16	6 to 29	−3 to 40	24	7 to 43	−7 to 59
Total AIS	7	−11 to 32	−27 to 55	14	−12 to 56	−38 to 95	18	−14 to 67	−56 to 144
Total ice sheet	23	1 to 50	−16 to 76	31	2 to 76	−25 to 119	42	5 to 97	−34 to 181
2200									
EAIS	2	−19 to 23	−37 to 47	12	−14 to 43	−38 to 83	30	−17 to 206	−48 to 313
WAIS	18	−7 to 111	−32 to 161	37	0 to 137	−35 to 202	61	12 to 197	−31 to 313
GrIS	35	6 to 74	−11 to 101	58	21 to 108	−2 to 146	75	31 to 144	−1 to 204
Total AIS	26	−20 to 122	−59 to 195	52	−8 to 166	−66 to 272	114	4 to 365	−65 to 603
Total ice sheet	67	6 to 168	−34 to 251	118	37 to 242	−19 to 363	203	66 to 470	−11 to 727
2300									
EAIS	21	−25 to 97	−52 to 153	39	−18 to 166	−55 to 244	69	−20 to 296	−84 to 505
WAIS	34	−13 to 117	−55 to 174	85	14 to 215	−36 to 316	149	35 to 305	−33 to 457
GrIS	65	9 to 137	−21 to 191	84	22 to 167	−13 to 230	118	39 to 221	−7 to 301
Total AIS	59	−27 to 193	−92 to 307	134	12 to 346	−70 to 524	235	38 to 559	−81 to 930
Total ice sheet	131	23 to 285	−46 to 420	230	78 to 461	−7 to 664	365	137 to 719	10 to 1120

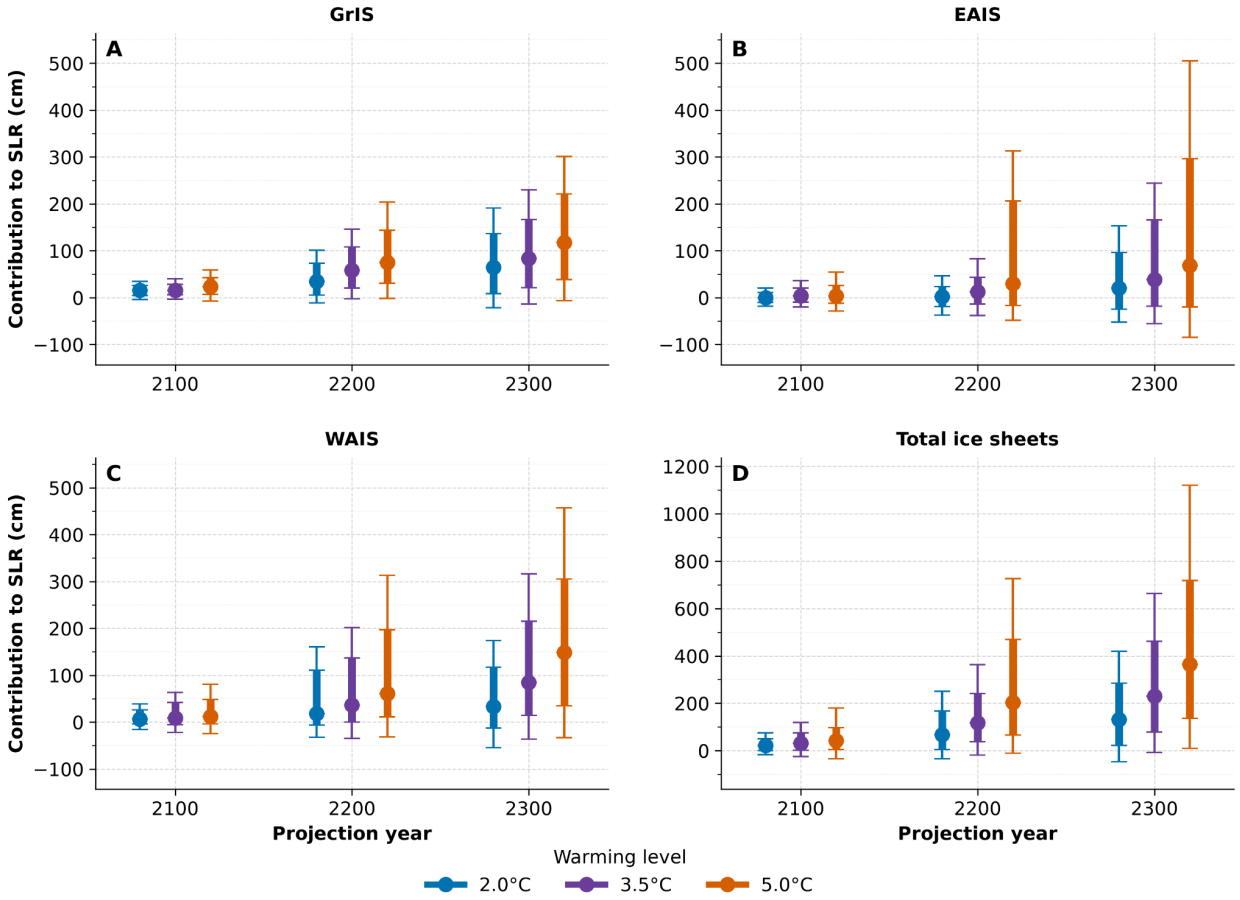
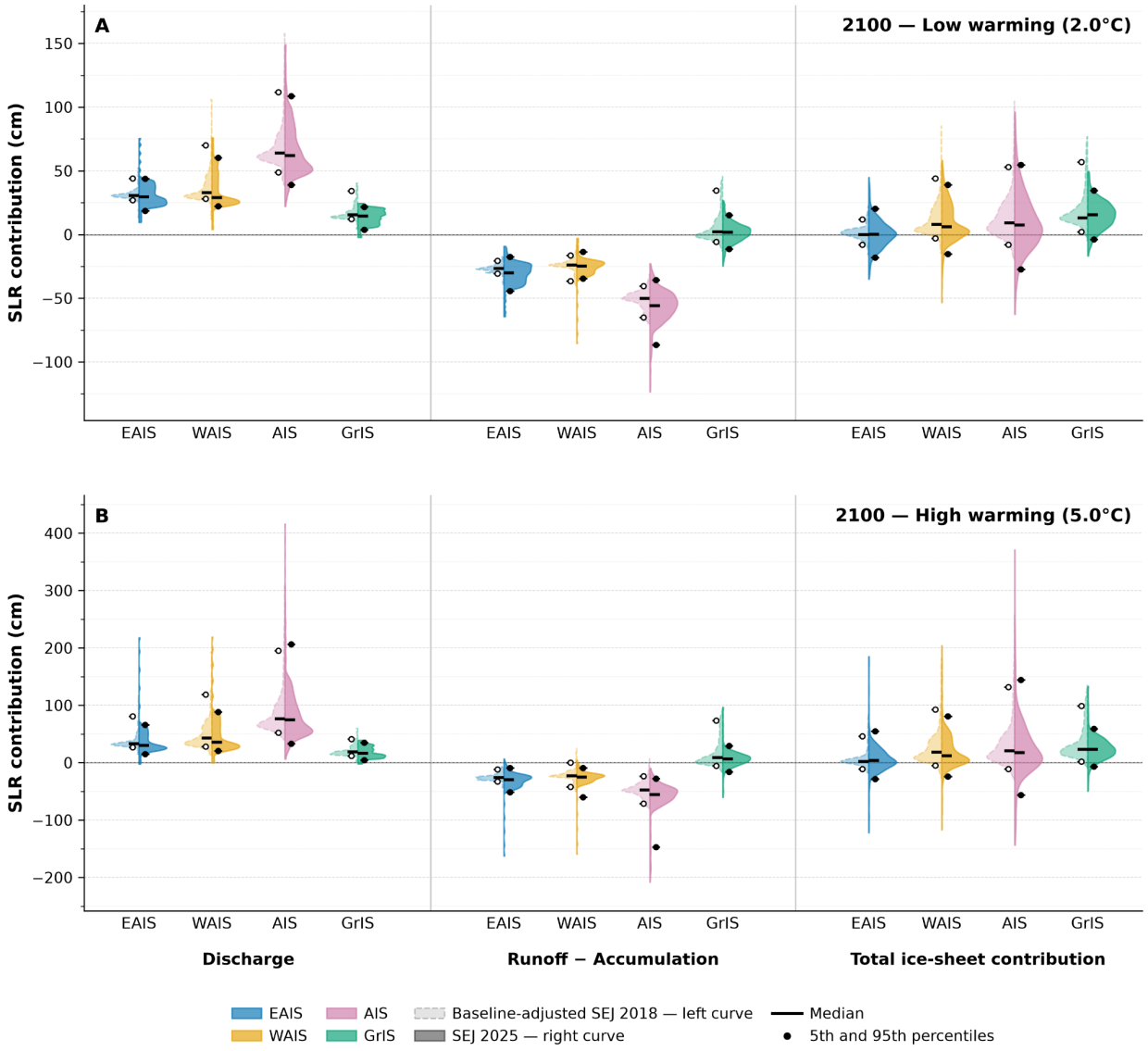


Figure 3. Projections of ice-sheet contributions to SLR under different temperature scenarios in 2100, 2200, and 2300. Panels show a) GrIS, b) EAIS, c) WAIS, and d) all the ice sheets combined. Bars are clustered by year and colored by warming level; thin and thick vertical lines indicate the 5th–95th and 17th–83rd percentile ranges, respectively, and circles indicate medians.

4.2 Relationship between ice-sheet processes and warming level

For GrIS, SMB is responsible for both the largest share of projection uncertainty and the primary source of scenario dependence, with a strong warming dependence in both median and 95th percentile estimates (Tables 2 and S14; Figures 4, 5). By contrast, experts expect a fairly constant rate of discharge, at a median estimate close to the baseline rate of about 1.4 mm/yr for the next three centuries, with limited sensitivity to warming trajectory. This is consistent with modeling studies that indicate that SMB dominates GrIS uncertainty for a fixed future climate (Aschwanden et al., 2019).

For AIS, temperature dependence emerges more slowly (Table 2 and S14). In 2100, median projections of both discharge and SMB exhibit minimal scenario dependence (Figure 4). Warming leads to a broadening of uncertainty in SMB at both ends of the distribution. Estimates of the 95th percentile of discharge in both WAIS and EAIS increase between 2.0°C and 3.5°C, with little change between 3.5°C and 5°C, suggesting expectation on the part of the experts of threshold behavior. In 2200, median SMB projections are likewise fairly stable across warming levels, with uncertainty expanding in both directions between 2°C and 5°C. 2200 projections for discharge do exhibit a clear increasing relationship with warming level, most strongly in 95th percentile projections for EAIS and in both median and 95th percentile projections for WAIS. By 2300, warming-level dependence emerges for SMB in median as well as 95th percentile for both EAIS and WAIS, and grows even stronger for discharge (Figure 5).



411

412 **Figure 4.** Probability distribution plots for SLR contributions by 2100 in the (A) Low (2.0°C) and
413 (B) High (5.0°C) warming scenario from the East Antarctic (EAIS), West Antarctic (WAIS), total
414 Antarctic (AIS) and Greenland (GrIS) ice sheets for discharge and surface mass balance (i.e.,
415 runoff minus accumulation), as well as in total. Results are derived from the SEJ for the 2018
416 (left curve) and current (right curve) elicitations. The 5th, 95th, and 50th percentile values of
417 the distribution are indicated by the circle and dash markers, respectively. The 2018 values
418 were adjusted to incorporate the same baseline trends provided to the experts in the SEJ 2025
419 structured-expert judgement elicitation.

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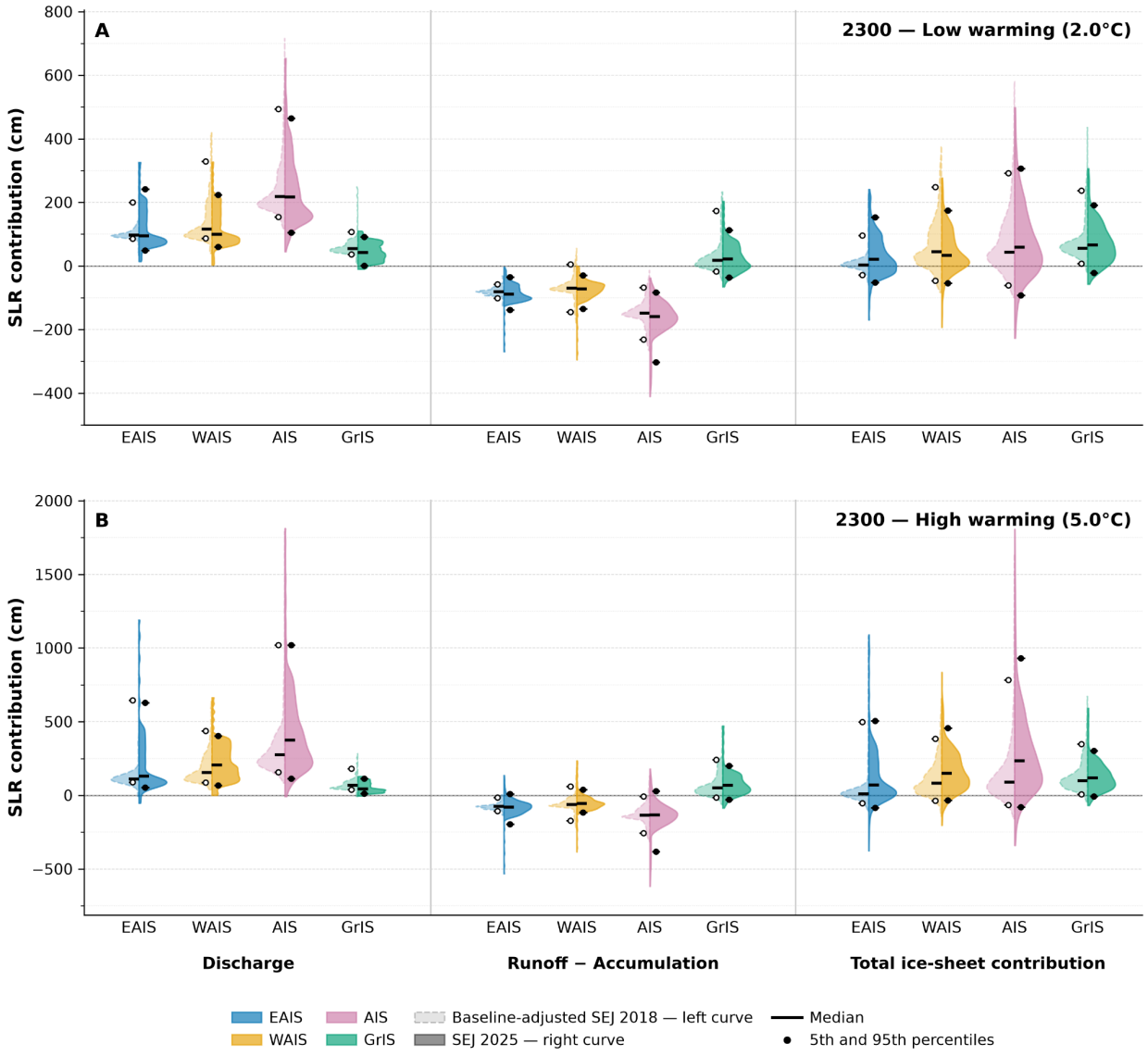


Figure 5. Probability distribution plots for SLR contributions by 2300 in the (A) Low (2.0°C) and (B) High (5.0°C) warming scenario from the East Antarctic (EAIS), West Antarctic (WAIS), total Antarctic (AIS) and Greenland (GrIS) ice sheets for discharge and surface mass balance (i.e., runoff minus accumulation), as well as in total. Results are derived from the SEJ for the 2018 (left curve) and current (right curve) elicitations. The 5th, 95th, and 50th percentile values of the distribution are indicated by the circle and dash markers, respectively. The 2018 values were adjusted to incorporate the same baseline trends provided to the experts in the 2025 structured-expert judgement elicitation.

4.3 Evolution of expert judgement since the 2018 SEJ

The judgement of the experts participating in the 2025 SEJ differed in several respects from the judgement of experts participating seven years prior in the 2018 SEJ (Figures 4, 5, and S4–S7).

While accumulation estimates for GrIS are similar between the 2018 and 2025 SEJs, the current experts give greater weight to the probability of a significant increase in accumulation in AIS. For example, for 2100 under the 5°C scenarios, the current experts estimate a 5th–95th percentile accumulation contribution to GMSL of −50 to −18 cm from EAIS and −66 to −20 cm for WAIS; this compares to the 2018 SEJ estimates of −33 to −12 cm for EAIS and −43 to −14 cm for WAIS (Figure 4, Table S15). By 2300 in the 5°C scenario, the difference for EAIS is even more stark: −239 to −52 cm, compared to the 2018 SEJ estimates of −106 to −16 cm. The same is not true for WAIS, however, where uncertainty in the accumulation contribution shrinks from −187 to −10 cm in the 2018 SEJ to −103 to −26 cm herein (Figure 5). While it is unclear what the precise cause is for this shift in judgement between 2018 and 2025, we note that in recent years the EAIS has experienced a significant increase in mass due to net accumulation from 2021–2025 (Kolbe et al., 2026; Stokes et al., 2025).

Though median estimates of runoff from GrIS are almost identical in this study and the 2018 SEJ, they are more precise in this study. For 2100, the 2018 SEJ projected a contribution of 18 (13 to 49) cm under 2°C and 24 (13 to 87) cm under 5°C; in the current study, experts project 19 (10 to 32) cm under 2°C and 26 (11 to 38) cm under 5°C (Figure 4, Table S16).

Median projections of discharge remain similar between this study and the 2018 SEJ but high-end uncertainty is somewhat reduced. For 2100, 95th percentile discharge projections remain essentially unchanged at 44 cm from EAIS under 2°C but shrink for all other ice sheets: for WAIS and GrIS at 2°C, shrinking from 70 and 34 cm, respectively, to 60 and 22 cm; for EAIS, WAIS and GrIS at 5°C, shrinking from 81, 119, and 41 cm to 66, 88, and 35 cm, respectively (Figure 4, Table S15). This is not uniformly the case over all time, however; expert projections of high-end discharge from EAIS in the 2°C scenario approximately match 2018 SEJ projections in

2200 at 104 cm and exceed 2018 SEJ projections in 2300, growing to 242 cm compared to the 2018 SEJ estimate of 200 cm (Figure 5).

Aggregated across processes, projections for changes in AIS exhibit a notable degree of consistency between studies (Table S16), particularly at the high end. In 2100, AIS contributions in this study are −27 to 55 cm at 2°C and −56 to 144 cm at 5°C, compared to −8 to 53 cm and −11 to 132 cm in the prior study (Figure S4). GrIS projections have reduced uncertainty compared to the prior study: −4 to 35 cm at 2°C and −7 to 59 cm at 5°C, compared to 2 to 57 cm and 2 to 99 cm in the prior study (Figure S4).

In general, expert judgement of the high end of the potential contribution of ice sheets to GMSL rise has been remarkably stable over the seven years between the two SEJ studies, despite considerable debate in the literature around topics like MICI: 95th percentile projections of 76 m and 181 cm at 2°C and 5°C in 2100, growing to 420 cm and 1120 cm by 2300, compared respectively to 81, 178, 426, and 988 cm, respectively, in the 2018 SEJ – differences of ≤ 6 cm, except for the 5°C projection in 2300 (Figures 4, 5). Similarly, median estimates remain fairly stable except under strong forcing and longer timescales: 23 and 42 cm at 2°C and 5°C in 2100, growing to 131 and 365 cm in 2300, compared to 26, 51, 120, and 225 cm in the prior study — differences of less than 11 cm, except for the 5°C projection in 2300. This stability suggests that the upper end of the IPCC AR6 low confidence sea level projections, which incorporated information from the 2018 SEJ, remains a reasonable indicator of plausible high-end SLR, when considering low-agreement ice-sheet processes.

By contrast, thanks primarily to the revised assessment of AIS accumulation, the estimated low end is significantly lower: 5th percentile estimates of −16 cm and −34 cm at 2°C and 5°C in 2100, and of −46 cm and 10 cm in 2300, compared to prior estimates of 1, 1, 0, and 14 cm, respectively. In all cases, the fifth percentile has lowered substantially, becoming negative, with the exception of the 5°C projection in 2300 (Figures 4, 5).

Table 2. Projected sea-level rise (SLR) contributions (in cm) from each ice sheet due to discharge and surface mass balance (i.e., runoff – accumulation). Values are the 50th (5th–95th) percentiles of the Monte Carlo distributions. Values of projected SLR contributions from runoff and accumulation separately are presented in Table S14.

Year and warming	Discharge			Surface mass balance (Runoff + Accumulation)		
	EAIS	WAIS	GrIS	EAIS	WAIS	GrIS
2100						
2.0°C	29 (18 to 44)	29 (22 to 60)	14 (4 to 22)	−30 (−44 to −17)	−25 (−35 to −14)	2 (−11 to 15)
3.5°C	30 (21 to 68)	33 (19 to 82)	12 (9 to 23)	−28 (−45 to −12)	−25 (−54 to −11)	3 (−12 to 20)
5.0°C	30 (15 to 66)	36 (20 to 88)	16 (4 to 35)	−30 (−51 to −9)	−25 (−60 to −9)	7 (−16 to 29)
2200						
2.0°C	61 (37 to 104)	62 (45 to 215)	29 (5 to 50)	−62 (−94 to −32)	−51 (−74 to −27)	6 (−25 to 56)
3.5°C	66 (37 to 145)	77 (46 to 222)	27 (8 to 60)	−57 (−87 to −21)	−47 (−88 to −8)	28 (−18 to 94)
5.0°C	69 (30 to 346)	97 (54 to 287)	28 (8 to 84)	−59 (−103 to −7)	−46 (−78 to 10)	40 (−21 to 127)
2300						
2.0°C	94 (48 to 242)	100 (60 to 224)	42 (1 to 92)	−89 (−138 to −36)	−72 (−135 to −30)	22 (−36 to 113)
3.5°C	110 (49 to 335)	146 (71 to 328)	41 (10 to 106)	−83 (−139 to −24)	−65 (−118 to −1)	36 (−34 to 138)
5.0°C	131 (53 to 629)	206 (68 to 404)	44 (11 to 114)	−81 (−197 to 9)	−57 (−116 to 38)	67 (−30 to 201)

4.4 Implications for sea-level rise projections

Interpolating the expert-elicited ice-sheet projections to climate trajectories from CMIP6 and CMIP7, as described in Section 2.5, yields a substantially broader range of potential ice-sheet contributions to GMSL rise than represented in the IPCC AR6 likely range projections, which incorporate only ice-sheet processes whose projections are evaluated as being characterized by at least medium confidence (Figures 6, S8–11; Tables S17–28; Text S9). Under the CMIP6 scenarios (Figures S12—14), the median (5th–95th percentile) contribution from the combined GrIS and AIS in 2100 ranges from 23 (−17 to 80) cm under SSP1-1.9 and 28 (–21 to 109) cm under SSP2-4.5 to 37 (−27 to 149) cm under SSP5-8.5. The scenario range is more pronounced by 2300, where the total ice-sheet contribution reaches 136 (−46 to 463) cm under SSP1-1.9, 192 (−27 to 679) cm under SSP2-4.5, and 296 (−3 to 974) cm under SSP5-8.5. Under SSP5-3.4-OS, projected total ice-sheet contributions remain intermediate between SSP1-2.6 and SSP2-4.5, reaching 26 (−20 to 102) cm in 2100 and 169 (−36 to 630) cm by 2300.

Relative to the medium-confidence ice-sheet projections assessed in AR6, which are based on emulation of the ISMIP6 (Edwards et al., 2021; Goelzer et al., 2020; Seroussi et al., 2020) and LARMIP2 (Levermann et al., 2020) multi-model ensembles, the expert projections indicate substantially greater potential for high-end ice-sheet contributions to SLR. For example, under SSP5-8.5, the projected 2100 95th-percentile contributions from GrIS (55 cm) and AIS (113 cm) are more than double the 95th-percentile AR6 medium confidence projections (23 and 56 cm, respectively) (Fox-Kemper et al., 2021; Garner et al., 2021; Kopp, Garner, et al., 2023). By comparison, the AR6 low-confidence projections, which incorporated Bamber et al. (2019) as a line of information, yield more similar results for AIS (95th percentile projection in 2100 under SSP5-8.5 of 131 cm) and higher results for GrIS (99 cm).

Notably, by 2150, all scenarios, even SSP1-1.9, yield ice-sheet contributions to GMSL exceeding 1.0 m at the 83rd percentile and 1.8 m at the 95th percentile, with 83rd percentile projections exceeding 2.0 m under SSP5-8.5, and 95th percentile exceeding 2.5 m under SSP2-4.5 and higher emissions scenarios. Importantly, these numbers consider only the ice-sheet contribution, and do not reflect other contributions to GMSL, such as global-mean thermal expansion. They thus emphasize the real possibility of multiple meters of GMSL rise occurring before the middle of the 22nd century.

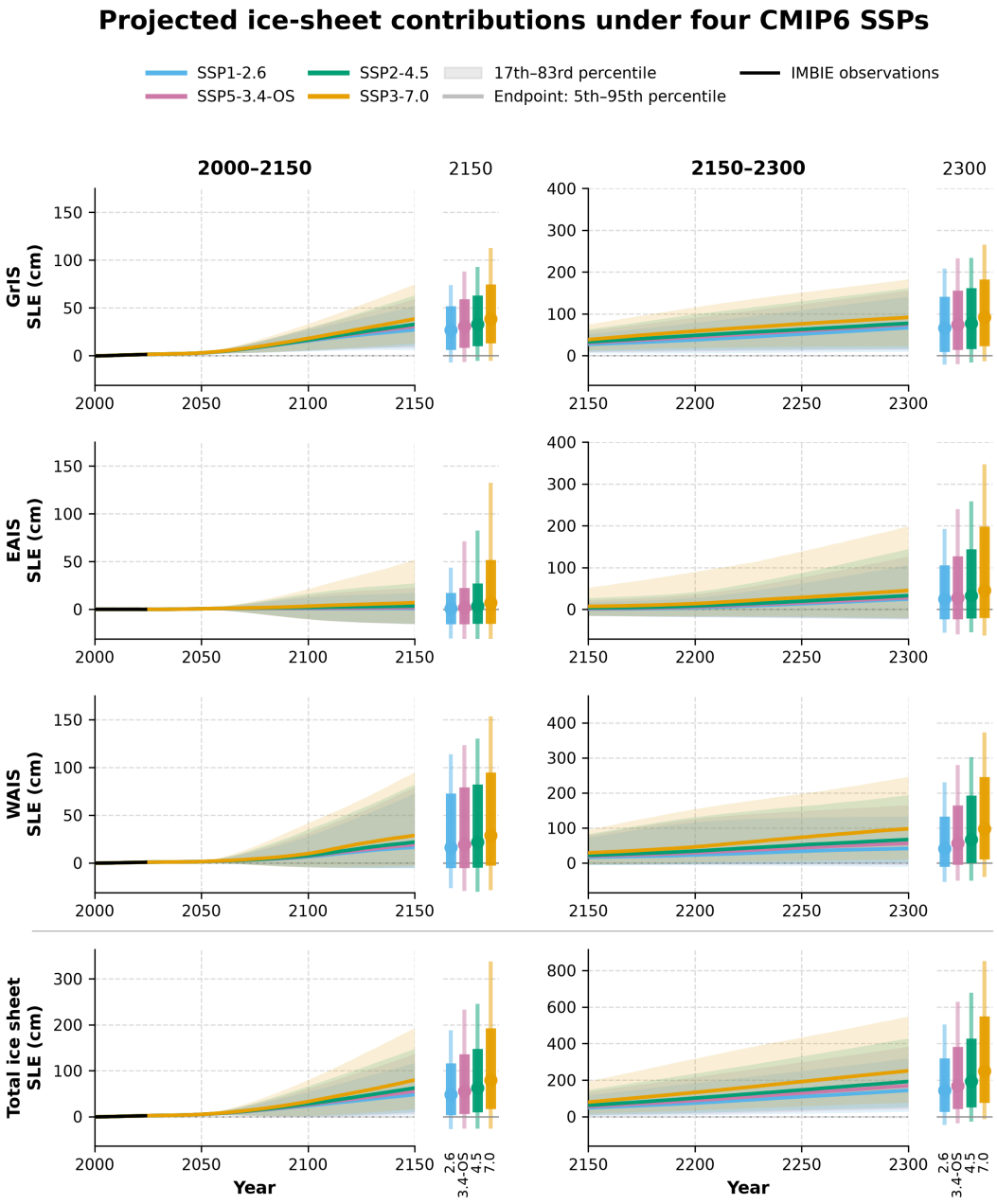


Figure 6. Projected ice-sheet contributions to global mean sea-level rise (SLR) from the Greenland Ice Sheet (GrIS), East Antarctic Ice Sheet (EAIS), and West Antarctic Ice Sheet (WAIS) under four CMIP6 scenarios from 2000–2300. The left and right columns show projections from 2000–2150 and 2150–2300, respectively, under SSP1-2.6, SSP5-3.4-OS, SSP2-4.5, and SSP3-7.0. Colored lines indicate median projections, with shaded bands representing the 17th–83rd percentile ranges. Adjacent vertical lines show the 5th–95th (thin lines) and 17th–83rd (thick bars) percentile ranges, with dots indicating medians, at 2150 and 2300. All contributions are

expressed as centimeters of sea-level equivalent relative to the 1995–2014 baseline. Note differences in y-axis scales. SLE = sea-level equivalent.

5 Expert reasoning, dissensus, and ambiguity

5.1 Expert dissensus and ambiguity in ice-sheet projections

The primary results in this paper represent the distributions of a "decision maker" who combines the estimates of different experts based on their accuracy and informativeness. In certain decision-making frameworks that do not require single probability distributions, other ways of looking at the results are also valuable.

For example, parallelling the (Lempert, 2002) definition of deep uncertainty, ambiguity arises when analysts do not know or cannot agree on the model structure and/or the distribution of key model parameters that is appropriate to describe a system. Dissensus among individual experts in a SEJ study thus relates closely to ambiguity. It is not identical to ambiguity, because some portion of the dissensus is reconciliable: The experts produced their responses individually, making different assumptions and implementation choices, but they might have been able to agree if there were a round of discussion and subsequent revision to their evaluations. Another portion reflects irreconcilable disagreements and genuine ambiguity – this portion would not be eliminated by subsequent discussion, because the experts fundamentally do not agree about key assumptions. Evaluation of experts' surveyed and narrative rationales, discussed below, provides further insights into irreconciliable disagreements.

Recognizing that the dissensus among individual experts reflects both reconciliable differences and ambiguity, we can nonetheless employ this dissensus as a proxy for ambiguity. Accordingly, to evaluate ambiguity in ice-sheet projections, we compare the spread in individual experts' 5th and 95th percentile responses to the mean estimate among these experts of the width of the 5th to 95th percentile range (Figures S18–44). (Henceforth, we refer to this quantity as the 5th-percentile or 95th-percentile dissensus-to-uncertainty ratio.)

At the 5th percentile, expert dissensus is greatest for the accumulation contribution to SLR (Figures S24–26, S33–35, S42–44). (Note that, because we elicited contributions to SLR, the

5th percentile of the accumulation contribution to SLR reflects the 95th percentile of mass accumulation at the ice sheet.) With the exception of GrIS accumulation in 2300, all accumulation estimates (and no discharge or runoff estimates) exhibit a 5th-percentile dissensus-to-uncertainty ratio greater than 2. In the Antarctic, dissensus increases with warming level. At the 95th percentile, dissensus is least for estimates of accumulation, with no accumulation variable having a 95th-percentile dissensus-to-uncertainty ratio greater than 1.74. Among accumulation variables, dissensus in the 95th percentile is highest for the Greenland ice sheet and generally increases over time. Dissensus regarding the 95th percentile of Antarctic accumulation is typically lower, and decreases with warming level.

Dissensus regarding 95th percentile of runoff is higher for AIS than for GrIS (e.g., Figures S18–20, S27–29, and S36–38), with all but one AIS runoff variable having a 95th-percentile dissensus-to-uncertainty ratio greater than 3. No GrIS discharge estimate has a 95th-percentile dissensus-to-uncertainty ratio greater than 2. The dissensus-to-uncertainty ratio for 95th percentile WAIS discharge projections for 2300 is relatively low (1.4–1.8) because of the broad uncertainty in expert estimates (mean range of 1.0–2.0 m). For earlier time periods, dissensus is substantial; for WAIS and EAIS discharge in 2100, the 95th-percentile dissensus-to-uncertainty ratio is greater than 4 (spread of 1.7 m for each ice sheet, compared to a width of 0.4 m).

5.2 Drivers of ice-sheet change

Compared to the 2010 and 2012 SEJs reported in Bamber and Aspinall (2013) and the 2018 SEJ (as reported in (Bamber et al., 2022)) , the experts in the 2025 SEJ are more certain regarding the role of anthropogenic forcing as the dominant driver of recent mass trends (since 1990) in GrIS and WAIS (Figure S55). For GrIS, 100% of experts identified anthropogenic forcing rather than internal variability as the dominant driver, compared to 90% in the 2018 SEJ. For WAIS, 67% identified anthropogenic forcing as dominant, compared to 50% in the 2018 SEJ. For the 2010 and 2012 survey the question was phrased differently and a direct comparison is not feasible, but the results suggested a lack of consensus in the trends for the WAIS and GrIS, with the median value lying at about 50% (Bamber & Aspinall, 2013).

584 Looking forward, experts judged ice dynamics to be the primary driver of GrIS
585 projections for 2100 under all scenarios, and for 2200 and 2300 under 2°C, with SMB
586 dominating GrIS changes in 2200 and 2300 under 3.5°C and 5°C scenarios (Figure 7). This is a
587 notable change from the 2018 SEJ, where experts judged SMB to dominate GrIS projections in
588 the 21st century as well. Among specific drivers named in the elicitation, the experts judged
589 basal traction to dominate dynamic processes for GrIS, with atmosphere circulation/moisture
590 transport dominating SMB.

591 In Antarctica, for 2°C of warming, expert opinion is closely split between ice dynamics
592 and SMB being the primary driver, with the primary dynamic driver being ice shelf buttressing
593 and the primary SMB driver being atmospheric circulation and moisture transport (Figure 7). Ice
594 dynamics become increasingly dominant with warming and with time. For 5°C of warming,
595 WAIS mass change in 2300 is dominated by factors associated with MISI and MICI: ice shelf
596 buttressing, ice shelf hydrofracturing, and ice cliff instability. While SMB is more significant for
597 EAIS than for WAIS, most experts judge EAIS to be dominated by ice dynamics by 2300 in the
598 5°C scenario. For both EAIS and WAIS, Ice shelf hydrofracturing and ice cliff instability are
599 judged to be the two largest drivers of ice dynamics in 2300, but experts also named ice shelf
600 buttressing, transverse stresses, and basal traction as key factors.

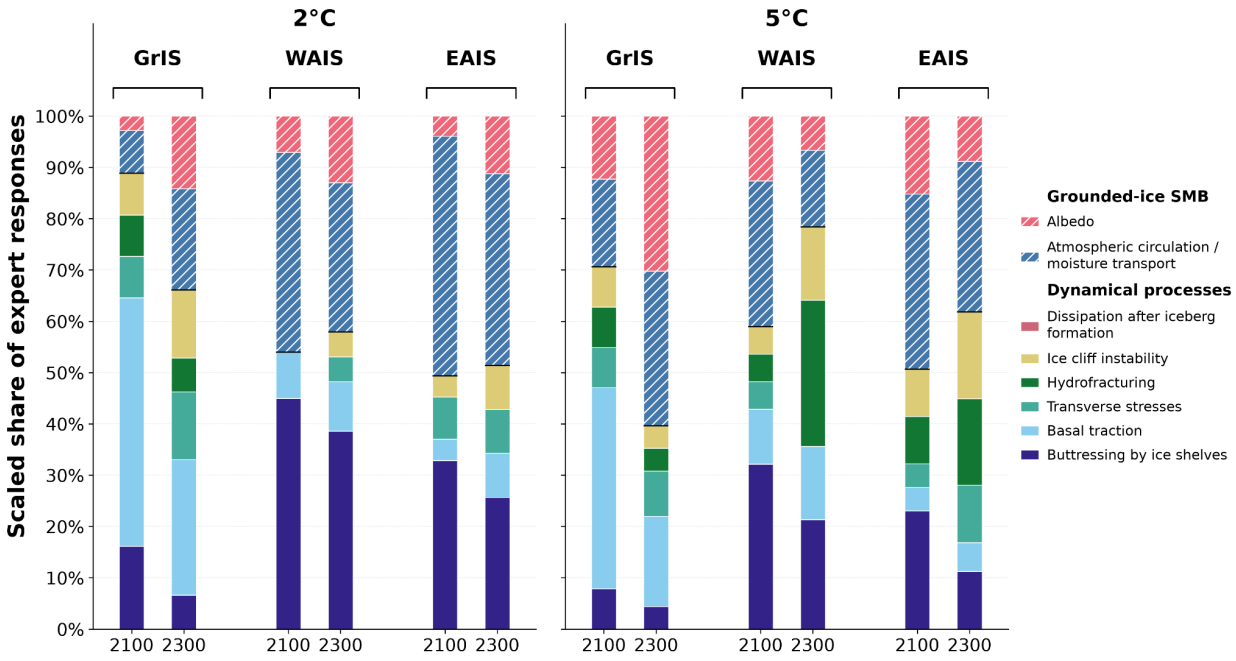


Figure 7. Expert judgements on the relative role for six drivers of changes in ice dynamics (i.e., buttressing by ice shelves, basal traction, transverse stresses, hydrofracturing, ice cliff instability, and dissipation after iceberg formation) and two drivers for changes in surface mass balance (SMB) for grounded ice (i.e., atmospheric circulation/moisture transport and albedo) at projection years 2100, 2200, and 2300 under 2.0°C and 5.0°C warming for the East Antarctic (EAIS), West Antarctic (WAIS) and Greenland (GrIS) ice sheets. Descriptive definitions for these factors are provided in Note S# in the supplemental. SMB processes were considered separately for grounded and floating ice. SMB and dynamical processes are scaled according to their relative contribution to the integrated ice sheet substantial sea-level rise. Note that, unlike in the numerical projections, expert responses are not weighted.

5.3 Narrative rationales

IPCC AR6 treated ice-sheet projections from SEJ as low confidence because "individual experts participating in the SEJ study may have incorporated processes in whose quantification there is low confidence, and the experts' reasoning has not been examined in detail" (Fox-Kemper et al., 2021). One innovation in this study is the inclusion of narrative rationales from 13 of the 21 participating experts, which allows their reasoning to be examined in greater detail. (The full rationales are provided in Supporting Information Text S6.)

The narrative rationales complement the formal questions in the elicitation, which ask the expert to identify dominant processes (Section 5.2). They allow the experts to convey more

nuanced reasoning and make the expert reasoning underlying the projections more transparent. The rationales highlight, for example, the widespread use of the Clausius-Clapeyron-type scaling relationships to inform estimates of accumulation (9 of 13 narratives) and consideration of MISI (9/13), as well as considerations such as topographic constraints on ice-sheet evolution (8/13) and the role of buttressing ice shelves in Antarctica (7/13), and the elevation/lapse rate feedback in Greenland (6/13). Other considerations mentioned in some narratives include the effects on surface mass balance of a shift from snowfall to liquid rain (6/13), albedo feedbacks in Greenland (4/13) and East Antarctica (1/13), and firn saturation thresholds (3/13), as well as the dynamic effects of changes in sub-ice shelf ocean circulation (3/13) and solid-Earth feedbacks in Antarctica (3/13).

The narrative rationales also highlight fundamental disagreements among experts regarding MICI. Three of 13 narratives explicitly include MICI, while six explicitly exclude it, due to either a judgement that its likelihood of significant contribution is low or because of an unwillingness to assess its likelihood given expert dissensus (Table 5). Relatedly, the rationales highlight the broad range of epistemic stances taken toward deep uncertainty: from explicitly linking deep uncertainty to low probability (Table 5, narr. 4), on one end, to refusing to attempt quantitative estimates in the presence of sufficiently deep uncertainty on the other end (Table 5, narr. 12). Notably, these two end-member stances correspond to the stands of the IPCC Third Assessment Report, which judged major 21st century loss of Antarctic grounded ice to be “very unlikely” and the IPCC AR4, which concluded that they could not assess the likelihood of scaled-up ice discharge in Antarctica, and therefore could not make a best estimate of 21st century SLR (Kopp et al., 2023).

Table 3. Statements in narrative rationales regarding Marine Ice Cliff Instability (MICI).

#	Statement
Narratives explicitly including MICI	
7	"[Discharge has] the greatest uncertainty. This is especially when considering key potential instabilities, especially MISI and MICI, although I understand the latter is very uncertain - and unlikely under low to medium emissions scenarios - and we cannot be sure that MICI exists in the real world. However, with high emissions warming scenarios, all bets are off the table. Discharge (perhaps alongside changing rates of basal melt, which are highly dependent on changes in ocean currents and mixing of different water masses under ice shelves), must contribute by far the greatest uncertainty for WAIS mass loss projections."
11	"Discharge will increase due to increased basal lubrication (a consequence of increased fracturing and subsurface runoff) and amplified ice-ocean interactions. Glaciers will find themselves in deeper/wider troughs, which may accelerate the discharge rate. Ice shelf collapse may trigger MICI in Thwaites, Jakobshavn, and other marine-based glaciers. Rising bedrock may slow down the progression of the grounding line retreat, mainly in WAIS, which is underlain by "soft" mantle. As temperatures stabilize, discharge may taper asymptotically in the 23rd century (recall the e-folding response of ice dynamics)."
12	"With the advent of strong MISI and possible MICI effects after 2200, and basal melting potentially extending to the eastern Ross and Filchner ice shelves, estimates for 2300 are unwarranted…. Dramatically increased SST, and the possibility of large-scale processes taking hold, make the long-range forecasting difficult. Among these are the AMOC shut-down, MISI, MICI, and new extents of mCDW [(modified Circumpolar Deep Water)] contact with the Antarctic coast. As experts, we can judge that some of these will come to pass; but the scale of the impact, the uncertainty as to when they will transpire, and the unknowable knock-on effects once they occur, make numerical estimates pointless. As an expert, all I want to say is 'these things are likely to occur if we continue on this course; and we don't want to venture there'."
Narratives explicitly excluding MICI	
1	"MICI is typically not accounted for by most physical models but, given that there is a lack of consensus within the ice sheet community about whether this process will be initiated, I did not adjust my judged projections to account for the possibility of MICI increasing discharge in the future."
4	"I have excluded deep uncertainties from my considerations because I define them as outside the probability range that was elicited, i.e. the 90% range around the median. These deep uncertainties include abrupt ocean circulation changes that strongly increase basal ice shelf melting around Antarctica and a runaway ice discharge through the Marine Ice Cliff Instability both in Greenland and Antarctica."
5	"MICI is not directly included, as the scenarios are climate stabilisation scenarios with a maximum of 5 degrees reached in 2100, with more recent findings indicating that MICI might not be relevant until 2100 (Morlighem et al., 2025; Rosier et al., 2025). However, I do account for the possibility of ice shelf collapse, and corresponding high rates of ice loss as identified in modelling studies (Seroussi et al., 2024)."
6	"Given the current debate on cliff failure, I did not account for the possibility of Marine Ice Cliff Instability as I do not think there is any observational evidence for such a mechanism. Marine Ice Sheet Instability, on the other hand, is well understood, observed, and properly captured by numerical models."
8	"I did not factor in MICI, for which there is little to no evidence."
10	"My approach was to rely on the published work based on time-variable gravity observations in Greenland and Antarctica, and reconstructions of surface mass balance, - RACMO - along with sensitivity studies of the SMB models and ice discharge to warming, also derived from observations and multiple sources. I did not factor feedbacks such as MICI."

Narratives 2, 3, 9, and 13 did not mention MICI.

5 Conclusions

Deep uncertainty in ice sheet projections has been a reality acknowledged by many in the scientific community for most of the last five decades, and remains a reality today. In the presence of deep uncertainty, some – like the IPCC AR4 – decline to evaluate the likelihood of future ice-sheet and sea-level changes. But that disinclination fails to meet the needs of decision makers and society at large, and SEJ offers an alternative, well-established approach for estimating the likelihood of future changes despite deep uncertainty.

This SEJ study confirms that experts give significant probability to outcomes outside the IPCC AR6's likely ranges. Those ranges incorporate only processes whose behavior is characterized by at least a medium level of evidence and a medium level of agreement among evidence. The SEJ results imply that, even under an intermediate-emissions trajectory comparable to that expected under current global policies (SSP2-4.5), the ice-sheet contribution to GMSL rise could at the 95th percentile reach 1.1 m by 2100, 2.5 m by 2150, 3.9 m by 2200, and 6.8 m by 2300. Although the experts express fundamental disagreements regarding MICI as a driver of ice-sheet change, their quantitative responses emphasize a significant, fat-tailed hazard associated particularly with Antarctic ice dynamics.

The 2018 SEJ of Bamber et al. (2019) provided a crucial input to the ice sheet and sea-level projections of IPCC AR6 that incorporated ice-sheet processes marked by limited evidence and/or a low level of agreement among evidence (Fox-Kemper et al., 2021). In particular, in AR6's 'low confidence' projections, the 2018 SEJ was the primary source of quantitative high-end constraints on Greenland evolution and one of two sources of high-end constraints on Antarctic evolution. The stability in 2100 median and 95th percentile projections for total ice-sheet contribution to GMSL between the 2018 SEJ and the present study suggest that the AR6 high-end estimates remain a robust indicator of the plausible range of future GMSL rise, when incorporating processes that are characterized by limited evidence and/or agreement and thus excluded from AR6's likely range projections.

We note, however, that the role of the AIS in the 90th percentile credible range for SLR has increased relative to the GrIS for all temperature trajectories at 2100 (Figure 4), suggesting

that recent research has reduced uncertainty for the GrIS while expanding it for AIS. This can appear counter-intuitive but it is not uncommon for uncertainties to increase with improved process understanding, sometimes described as “the expansion of the horizon of ignorance” (Roe & Baker, 2007). Our results provide a guide to where future research efforts should be focused.

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The expert participants-authors (SA, DF, EH, TH, AL, MM, RR, ER, TAS, HS, CRS, LDT, and IV) contributed resources (specifically, their time and knowledge) to this manuscript through the elicitation protocol and narrative rationales. The results and conclusions of this paper are based on application of the elicitation protocol and do not necessarily represent the views of individual expert participant-authors or the organizations with which they are affiliated. Text S10 further outlines the co-authorship standards for expert participants.

Open Research

Data Availability Statement

The SEJ elicitation data underlying this study, including anonymized expert assessments and the resulting performance-weighted distributions used to generate 50,000 Monte Carlo samples of EAIS, WAIS, and GrIS contributions to GMSL change, are available on Zenodo at https://doi.org/10.5281/zenodo.23543612. Historical ice-sheet mass-balance observations used to constrain the projections were obtained from IMBIE (Otosaka et al., 2026), available through the UK Polar Data Centre at https://doi.org/10.5285/128c5e33-5224-4197-82f0-19dcc95b80a0.

Software Availability Statement

SEJ processing software (Excalibur) and Bayesian network (Uninet) software for dependence modeling and sampling are freely available for non-commercial use from LightTwist Software at at https://lighttwist-software.com/excalibur/ and https://lighttwist-software.com/uninet/, respectively. The Framework for Assessing Changes To Sea-level v2 (FACTS2), used for generating temperature and emulated ice-sheet-model projections, is available at https://fact-sealevel.github.io/. Python scripts used to couple Monte Carlo samples temporally, construct annual ice-sheet contribution trajectories through 2300, apply the projections to CMIP6 and CMIP7 climate pathways, and produce the figures and tables presented in this study, are developed at https://github.com/bobkopp/ice-sheet-sej-2025 and will be deposited in Zenodo at time of acceptance.

Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

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